AI and Mathematical Progress: Truth, Understanding, and the Work Ahead

A response to *A Severe Misalignment of AI in Mathematics*

Bacim Alali
bacimalali@math.ksu.edu

The declaration published at mathandai.org raises serious concerns: mathematical work can be distorted by benchmark competition, premature announcements, inadequate attribution, and incentives that separate producing results from developing expertise. It also recognizes that AI can enhance mathematical understanding, that the profession must adapt, and that responsibility extends beyond technology companies. It insists that the purposes of mathematical work remain central as its means change.[1]

This response shares the declaration's concern for mathematical understanding, but questions whether solving important problems should be valued only as a means toward that goal. Establishing an important truth is itself a mathematical achievement, even when we do not yet have an illuminating explanation of why it holds. Moreover, knowing the answer to such a question can help researchers develop that explanation. Universities, journals, research funders, and companies that develop AI should therefore support both the discovery of reliable answers and the further work of understanding, interpreting, and applying them.

1. Mathematical value has more than one form

The declaration's description of solving problems as "only a tool and proxy" for conceptual understanding is too restrictive. On one reasonable reading, the phrase refers to the role famous problems have played as "landmarks and lighthouses" against which, in the declaration's words, one can measure an improved understanding of the mathematical landscape, and in that role it is apt: solving such a problem has often been the sign that new ideas had arrived. But even on that reading, an established answer can have mathematical value beyond the conceptual insight supplied by its first proof.

Several things can matter in a mathematical achievement: establishing whether a statement is true, introducing a method, revealing a structure, connecting previously separate ideas, enabling an application, and making an argument intelligible to others. These contributions often reinforce one another. They need not arrive together, and there is no universal ordering of their importance.

Sometimes a proof is more significant than the theorem it establishes because its methods open an entire field. Sometimes the result is more significant than the particular route by which it was obtained. A proof or counterexample that settles an important question may rule out a previously plausible theory or make further deductions possible. Its value need not wait for the discovery of a beautiful explanation.

Imagine a mathematically sound resolution of the question of whether there are infinitely many twin primes. We would want to know the answer even if the first proof were long, difficult, and conceptually unsatisfying. Our limited conceptual insight into why the result holds would give us further work to do; it would not make the established answer a mere stand-in for the achievement we actually sought. The speed or origin of the proof would not, by itself, diminish that achievement.

This does not place equal value on every true statement. Importance still requires judgment. It does mean that mathematical progress cannot be measured exclusively by either solved problems or explanatory insight. A serious account must accommodate both.

2. Knowing the answer can advance understanding

A reliably established answer can change what researchers investigate next. Before a conjecture is settled, researchers may be uncertain whether to seek a proof, construct a counterexample, or revise the conjecture. Once the answer is established, they can investigate why it holds, which hypotheses are essential, and whether a more

[1] *A Severe Misalignment of AI in Mathematics*, published September 11, 2026. mathandai.org; DOI: 10.5281/zenodo.22737750. Accessed September 21, 2026.

revealing argument exists. Those investigations may lead to new methods, generalizations, or even new areas of mathematics.

When a conjecture is proved, earlier results proved on the assumption that it is true no longer depend on that unproved conjecture; a proof of the Riemann Hypothesis, for instance, would at once make unconditional a large body of results in analytic number theory now stated under that assumption. Any other assumptions in those results still apply.[2] Useful intermediate results from the first proof can also become starting points for alternative approaches. Even a conceptually disappointing proof can identify a destination worth reaching by a better route.

The history of the prime number theorem provides a concrete example. It was proved analytically in 1896; the elementary proofs of Selberg and Erdős came half a century later, from mathematicians who knew exactly what they were aiming at, and showed that the theorem could be proved without the machinery of complex analysis.[3] For this later work, the question was no longer whether the theorem was true, but whether it could be proved by a different mathematical route. These later proofs illustrate how a settled theorem can remain the subject of original work on methods and mathematical understanding.

None of this guarantees that explanation will become easy. Knowing the answer does not tell us how much easier it will be to find an illuminating proof, and a first argument may offer few useful clues. Nevertheless, certainty about the mathematical target is a genuine resource. An answer can begin a new phase of research rather than conclude the subject. AI may participate in that phase too, helping to develop explanations, generalizations, and alternative arguments.

The declaration warns that "the mass production at faster and faster pace of 'true/false' statements could destroy fertile ground instead of breathing life into new ideas." The warning deserves to be taken seriously. Even a rapid succession of important, correct results can strain the community's capacity to check arguments, develop explanations, and educate new researchers. The value of establishing a result and the community's capacity to assess and develop a growing body of results are distinct questions. The pace of discovery affects the demands on that capacity; it does not, by itself, diminish the value of an individual discovery. Section 5 outlines a constructive response to these pressures.

The qualification is essential: the answer must be established. A plausible assertion or unexamined announcement does not settle the question. It may suggest a useful direction, but treating it as established can redirect effort toward a false premise. Accepting the answer requires a sound proof of the stated result, with its assumptions and reliance on earlier results made clear. Formal verification can provide a rigorous basis for accepting a proof, provided the formal statement faithfully represents the mathematical question and the assumptions and checking process are sound. Determining whether the formal definitions and hypotheses capture the intended question requires mathematical expertise; a successful formal check does not settle that issue. Reliability, not the manner or speed of discovery or the elegance of the argument, is what allows the answer to guide further inquiry.

3. The pressures extend beyond the AI companies

The practices of the AI companies developing frontier systems matter. Their choices about benchmarks, public announcements, access, and research support can influence what receives attention and how mathematical achievements are presented. Asking these companies to improve those practices is reasonable and potentially consequential.

But it cannot be the whole response. Whenever AI tools are accessible beyond the companies that develop them, mathematicians and other users can employ them to generate and circulate candidate results. A decision by those companies to stop announcing mathematical results would not, by itself, determine how other users conduct or communicate their work.

Priority competition, incomplete checking, inadequate attribution, and pressure on readers can arise across many institutions and individual workflows. Their severity depends on the capabilities and accessibility of the systems, corporate incentives and policies, and the standards researchers follow when checking, crediting, and sharing results. These influences interact: company practices can shape researchers' incentives, while community standards can influence what companies are expected to provide.

[2] For the significance of the Riemann Hypothesis and its consequences for the distribution of primes, see Enrico Bombieri, *Problems of the Millennium: the Riemann Hypothesis*, official problem description, Clay Mathematics Institute.

[3] Dorian Goldfeld, *The Elementary Proof of the Prime Number Theorem: An Historical Perspective*, in D. Chudnovsky, G. Chudnovsky, and M. Nathanson (eds.), *Number Theory*, Springer, 2004, pp. 179–192. Goldfeld reconstructs the development from the original papers and correspondence.

The practical issue is the balance between producing candidate results and assessing them. Wherever AI makes it easier to generate mathematical arguments without comparable improvements in checking, explaining, and organizing them, the burden on readers and reviewers can grow. AI can also assist with these other activities. The task is to develop and evaluate those uses deliberately, rather than assume that faster production will automatically bring equally effective verification and explanation.

The declaration itself calls for action by the mathematical community as well as by the companies developing AI. That response must address AI-assisted mathematics wherever it is produced, rather than rely primarily on changes within a few AI companies.

4. Existing practices also deserve examination

The declaration is right that discussion, sustained study, collaboration, exposition, and teaching take time, and right about what that time produces. Working through examples, seeing why one definition is the useful one, and learning to adapt a difficult argument are how mathematicians acquire the judgment to extend a result, to recognize when something has gone wrong, and to teach the subject to those who come next. Access to a proof does not by itself confer the judgment to adapt it or initiate a new inquiry. Better instruction, AI assistance, and study of established results may help that judgment develop more efficiently, but its development still requires active mathematical participation.

Yet none of this establishes that our current arrangements support these activities as well as they could. Published proofs may leave readers considerable work to identify dependencies and connections to other mathematics. A finished paper may omit why a natural approach failed, even when that failure reveals the obstacle the successful proof overcomes. Related results can remain disconnected because researchers in different mathematical fields use different terminology or seldom encounter one another's work. Opportunities to attract specialist attention and receive informed criticism also differ across institutions and professional networks.

The declaration's silence about these limitations does not mean that it considers the existing system perfect. But the mathematical community's response to AI should examine them explicitly. Journals, preprint repositories, seminars, conferences, and informal exchanges are the means through which researchers assess and learn about new work. These channels will face greater demands as the volume of results grows.

The incentive problem also predates AI. Counts of papers, claims of priority, and prestige of venue can compete with the work of clarification and synthesis. To be sure, alternative proofs and exposition sometimes receive great recognition. But hiring, promotion, funding, and publication decisions that emphasize new theorems can undervalue work that verifies an existing result, explains it more clearly, or connects it to other ideas. Such contributions need suitable publication venues and explicit recognition in research evaluation.

The objective should be to preserve and strengthen mathematical learning while improving how knowledge is communicated and assessed. Time spent working through a difficult proof is not the same as time lost because relevant work is hard to locate, essential explanations are missing, or knowledgeable feedback is difficult to obtain. Respecting the time needed for understanding does not require accepting these avoidable obstacles in publication, exposition, and access to expertise.

5. A constructive program

The first priority is to support public announcements of mathematical results with evidence that others can examine. An announcement that a problem has been solved should normally be accompanied by an accessible proof, whether a written argument or a machine-checkable formal proof, together with a precise statement of the result, its relevant assumptions, and an honest description of what has been checked. Preliminary claims should be identified as preliminary. Prior work and substantive contributions should be acknowledged, including contributions that do not fit neatly into conventional authorship.

These expectations should apply to AI companies, universities, and individual researchers alike. They do not require every informal argument to be formally verified or every exploratory calculation to become a publication. They require the strength of a public claim to match the evidence made available for it.

When a conclusion depends on computation, researchers should provide the code, inputs, relevant software versions, and computational conditions needed to examine or reproduce that part of the argument. If an AI system searches through many unsuccessful approaches before producing a proof that can be checked on its own, verifying the theorem does not require repeating that search. If a computation remains an essential step in the proof, however, that step must itself be justified. Reports of proofs discovered with substantial computing

resources should therefore distinguish the resources used for discovery from those needed for verification. Claims about the efficiency or reproducibility of the discovery method require their own supporting evidence.

The second priority is to make verification and explanation sustained research activities. Universities, AI companies, and research funders should support checking, exposition, and reusable verification tools alongside the production of new results. Part of this work requires better infrastructure for mathematical knowledge. In separate work, I proposed a system that records results together with their dependencies, helps identify when a claimed result is already known, and makes the methods within proofs easier to find and reuse.[4] The declaration raises a related concern when it observes that rushed announcements leave no time for "the isolation of new methods and ideas, and citing relevant previous work of others." Existing practices already support these activities unevenly. Strengthening them as the volume of mathematical work grows will require sustained investment in tools, expertise, and institutions, not appeals to better practice alone. Expert review of written arguments and formal proof checking should be supported as complementary activities. Formal-verification workflows should include expert review of the statement and assumptions, and make clear which logical foundations and trusted checking software the verification relies on.[5] Agreement among similar AI systems is not, by itself, independent verification: the systems may repeat the same error.

The third priority is recognition. Departments, journals, and funding agencies should assess the mathematical contribution of verification, illuminating proofs, unification, and exposition, rather than treating their value as derivative of first discovery. This need not become another volume-based metric. The relevant question is what the work enables others to know, understand, or do. As obtaining first proofs becomes easier, recognizing work that deepens understanding becomes more important, not less.

Participation also requires access. Universities and funding agencies should support shared computational resources, usable tools, and expertise in verification, including for researchers outside well-resourced institutions. The community's ability to shape mathematical AI will depend partly on its capacity to use, evaluate, and improve the systems on which it relies.

Finally, education must preserve opportunities to develop mathematical judgment. As argued in my work on graduate mathematical formation, understanding a supplied proof does not by itself demonstrate the capacity to initiate and direct research.[6] Learning to use AI effectively includes questioning its suggestions, testing examples, inspecting assumptions, and understanding arguments deeply enough to make responsible decisions. Students also need opportunities to formulate problems, choose approaches, and create mathematics, not merely supervise output. Some tasks should continue to be undertaken without AI assistance when independent work serves their educational purpose. Such choices need not limit the tools available for research.

6. Continuing the mathematical project

The declaration rightly puts the purposes of mathematics at the center: developing understanding, sustaining the people capable of it, and finding out what is true. A rapid, reliable result and the explanation that follows can both be original and consequential. Our institutions should support and recognize both achievements, rather than make one a substitute for the other.

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[4] Bacim Alali, *Toward a New Infrastructure for Mathematical Knowledge and Discovery*, manuscript, 2026.

[5] See *The Lean Language Reference*, *Validating a Lean Proof*: `https://lean-lang.org/doc/reference/latest/ValidatingProofs/`.

[6] Bacim Alali, *Graduate Mathematics in the Age of AI: Forming Mathematicians for Original, Independent, and Responsible Inquiry*, 2026, arXiv:2609.22891, `https://arxiv.org/abs/2609.22891`.